

HDOC Day 22 - Answer

Algorithm

1. Start. 2. Read positive m and n with $m \leq n$. 3. Display the menu repeatedly. 4. Choice 1 prints multiplication tables of m and n side by side for multipliers 1 to 10. 5. Choice 2 prints a multiplication grid for rows 1 to m and columns 1 to n. 6. Choice 3 prints each integer from m to n followed by its next five consecutive integers. 7. Choice 4 exits. 8. Use switch-case and show an error for invalid choices.

C Program

```
#include <stdio.h>

int main() {
    int m, n, choice, i, j;

    do {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d%d", &m, &n);
        if (m <= 0 || n <= 0 || m > n)
            printf("Invalid input. Try again.\n");
    } while (m <= 0 || n <= 0 || m > n);

    do {
        printf("\n1. Tables of m and n side by side\n");
        printf("2. Multiplication grid\n");
        printf("3. Next five consecutive integers\n");
        printf("4. Exit\nEnter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                for (i = 1; i <= 10; i++)
                    printf("%d x %d = %-4d    %d x %d = %d\n",
                           m, i, m*i, n, i, n*i);
                break;
            case 2:
                printf("    ");
                for (j = 1; j <= n; j++) printf("%4d", j);
                printf("\n");
                for (i = 1; i <= m; i++) {
                    printf("%4d", i);
                    for (j = 1; j <= n; j++) printf("%4d", i*j);
                    printf("\n");
                }
                break;
            case 3:
                for (i = m; i <= n; i++) {
                    for (j = 0; j <= 5; j++) printf("%d ", i+j);
                    printf("\n");
                }
                break;
            case 4: printf("Exiting program...\n"); break;
            default: printf("Invalid menu choice.\n");
        }
    } while (choice != 4);
    return 0;
}
```

Test-case table

Test	Input	Choice	Expected result
1	m=2, n=4	1	Displays tables of 2 and 4 side by side from x1 to x10.
2	m=2, n=3	2	Grid contains products for rows 1-2 and columns 1-3.
3	m=3, n=4	3	Rows: 3 4 5 6 7 8 and 4 5 6 7 8 9.
4	m=0, n=3	-	Invalid input; asks again.

Test	Input	Choice	Expected result
5	m=1, n=2	8	Invalid menu choice; menu continues.